

Steven Boada, Ph.D

Basic Information	(615) 200-0119 stevenboada@gmail.com	github.com/boada linkedin.com/in/theboada
Professional Experience	Activision Publishing, Inc. <i>Senior Data Scientist, Gameplay</i> As part of the game science team, I work to make Call of Duty better through data. In this role, I: • Partner with senior design leadership and cross-functional teams, including engineering and product management, to define key metrics, identify opportunities for game and business improvements, and drive data-informed decision making across the game. • Lead initiatives to understand how gameplay systems (maps, modes, weapons, in-game communication) influence top-line KPIs and suggest design improvements based on data insights. • Design, implement, and monitor experiments, A/B tests, and other methodologies to drive improvements in gameplay features and player engagement. • Drive data quality and ensure instrumentation across the game to capture accurate and actionable data; collaborate closely with engineering teams to ensure seamless data flow. • Lead the development and implementation of data models, production metrics, and visualizations to provide insights into player behavior, gameplay patterns, and overall game performance. • Manage and mentor junior data scientists, offering guidance in study design, statistical analysis, and effective data communication. <i>Senior Machine Learning Engineer</i> • Designed, built, and maintained ML infrastructure and tools, including autoML, CI/CD (Jenkins, GitHub Actions), model tracking (MLflow), and orchestration (Airflow) to support data-driven decision-making. • Crafted and implemented policy for external data ingestion and retention to ensure accessibility and proper data stewardship across the organization. • Developed near-realtime applications (via Spark streaming) to identify and mitigate in-game cheating and malicious behaviors, achieving action times in the low 10s of seconds. • Spearheaded the transition of ML infrastructure from AWS to GCP/GCS, optimizing scalability and performance. • Created machine learning models focused on customer conversion, churn prediction, and behavioral segmentation using survival analysis, clustering, and tree-based techniques. • Managed the day to day of a small team of junior data scientists to design, develop, and deploy recommendation systems that were integrated into Modern Warfare 2 (2022) and Modern Warfare 3 (2023). <i>Insight Data Science Fellow</i> • Helped optimize the way NYC health inspectors perform restaurant inspections in order to reduce the time critical health violations remain unaddressed. • Trained a random forest in Python to prioritize NYC restaurant inspections based on environmental variables and their past inspection histories and provided the results to NYC through an API deployed on AWS. • Resulted in NYC inspectors identifying ~2.5% more violations in the first half of an inspection window, leading to critical violations being discovered up to 7 days earlier than by the current approach implemented by NYC. <i>Dept. of Physics and Astronomy, Rutgers University</i> <i>Postdoctoral Research Associate</i> • Designed and built parallelized data processing pipelines to analyze TBs of astronomical imaging data, producing standardized catalogs and rigorous results that contributed to 2 peer-reviewed publications and several hundred hours of telescope time. • Collaborated with senior scientists and graduate students to ensure the quality of data products and oversee scientific analysis. • Contributed to open-source Python projects focused on astronomy, adding features and fixing bugs in tools like photometrypipeline, astLib, and easyGalaxy.	Boulder, Colorado January, 2024 – Present January, 2021 – 2024 New York, New York January, 2020 – 2021 New Brunswick, New Jersey September, 2016 – 2021

Skills	Machine Learning: Linear Models, Decision Trees, SVM, Clustering, Deep Learning, Survival Analysis CI/CD: Jenkins, Airflow, Docker Software and Computing: Open Source Contributor, Python, DataBricks, MLFlow, SQL, AWS/GCP, Spark, and other cloud computing applications Leadership: Experience organizing and leading workshops and collaboration meetings, Teaching and mentoring junior team members, Eagle Scout.	
Education	Texas A&M University, College Station, Texas ● Ph.D, Physics (astronomy focus), 2016	The University of Tennessee, Knoxville, Tennessee ● M.S., Physics (astronomy focus), 2009 ● B.S., Physics, 2007